UNIT I

INTRODUCTION OF WEB

Topics Covered

1.1 History of WWW
1.2 Role of Web Browser and Web Server
1.3 Client-Side Programming
1.4 IDE Applications of HTML
1.5 Web Protocols – HTTP and FTP

1.1 HISTORY OF WWW

What is WWW?

WWW stands for **World Wide Web**. It is a system of interconnected web pages and resources that can be accessed through the Internet using web browsers.

The Web allows users to access and view different types of information such as:

- Text
- Images
- Audio
- Video
- Documents
- Interactive web applications

Web pages are connected to each other using **hyperlinks**. A user can click a hyperlink to move from one web page or resource to another.

Origin of the World Wide Web

The World Wide Web was proposed by **Tim Berners-Lee**, a British computer scientist, while he was working at **CERN (European Organization for Nuclear Research)** in Switzerland.

His main goal was to create a system that would allow researchers to easily share and access information between different computers.

The main technologies associated with the Web include:

- **HTML (HyperText Markup Language)** – Used to create the structure of web pages.
- **HTTP (HyperText Transfer Protocol)** – Used for communication between web browsers and web servers.
- **URL (Uniform Resource Locator)** – Used to identify and locate resources on the Web.

Important Milestones in the History of WWW

1989

Tim Berners-Lee proposed the idea of a global hypertext system for sharing information.

1990

The first web server and web browser were developed.

The first web browser was called **WorldWideWeb**.

1991

The World Wide Web became available outside CERN, and the first website was introduced.

1993

CERN made the Web technology available to the public without royalty fees. This helped the World Wide Web grow rapidly.

1994

The **World Wide Web Consortium (W3C)** was founded to develop and maintain open standards for the Web.

1990s Onwards

Graphical web browsers such as **Mosaic** and **Netscape** helped make the Web popular among ordinary users.

Present Day

The Web has evolved from simple static web pages into advanced systems such as:

- Dynamic websites
- Web applications
- Social media platforms
- E-commerce websites
- Cloud services
- Online banking
- Video streaming platforms
- Online education systems

Basic Components of WWW

Component	Meaning
Web Browser	Software used by users to access and view web pages.
Web Server	Computer/software that stores and delivers web resources to clients.
Web Page	A document displayed in a browser, commonly created using HTML.
Website	A collection of related web pages and resources.

Component	Meaning
URL	The address used to locate a resource on the Web.
HTTP/HTTPS	Protocols used for communication between browsers and web servers.
Hyperlink	A clickable connection that takes users to another resource.

1.2 ROLE OF WEB BROWSER AND WEB SERVER

Web Browser

A **web browser** is a client-side software application that allows users to access, retrieve, interpret, and display resources available on the World Wide Web.

Examples of web browsers include:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge
- Apple Safari
- Opera

Functions of a Web Browser

1. Accepts URL

The browser accepts a URL entered by the user to identify the required web resource.

2. Sends Requests

The browser sends requests to web servers using protocols such as HTTP or HTTPS.

3. Receives Web Resources

The browser receives resources such as:

- HTML files
- CSS files
- JavaScript files
- Images
- Videos
- Documents

4. Interprets Web Pages

The browser interprets HTML and CSS and executes JavaScript to display web pages.

5. Provides Navigation

Web browsers provide navigation features such as:

- Back
- Forward
- Refresh
- Bookmarks
- History

6. Manages Storage

Browsers manage information such as:

- Cookies
- Cached files
- Local storage

7. Provides Security

Modern browsers provide security features such as:

- HTTPS support
- Website permissions
- Sandboxing
- Warnings about unsafe websites

Web Server

A **web server** is a computer system and/or software that stores website resources and responds to requests received from web clients such as web browsers.

Common web server software includes:

- Apache HTTP Server
- Nginx
- Microsoft IIS
- LiteSpeed

Functions of a Web Server

1. Stores Web Resources

A web server stores web pages and other website resources.

2. Receives Client Requests

The server listens for requests sent by web browsers and other clients.

3. Processes Requests

The web server processes incoming requests and determines the appropriate response.

4. Delivers Web Resources

It delivers resources such as:

- HTML
- CSS
- JavaScript
- Images
- Documents
- Other files

5. Supports Dynamic Content

A web server can communicate with application software and databases to generate dynamic web content.

6. Handles Multiple Requests

A web server can handle requests from many clients at the same time.

Difference Between Web Browser and Web Server

Web Browser	Web Server
Runs mainly on the user's device.	Runs on a server computer or hosting infrastructure.
Acts as a client.	Provides services to clients.
Requests and displays web resources.	Stores, processes, and delivers web resources.
Examples: Chrome, Firefox, Edge.	Examples: Apache, Nginx, IIS.

How Web Browser and Web Server Work Together

The communication between a browser and server generally occurs as follows:

Step 1: User Enters URL

The user enters a URL into the web browser.

Step 2: Browser Sends Request

The browser sends a request to the appropriate web server.

Step 3: Server Processes Request

The web server receives and processes the request.

Step 4: Server Sends Response

The server sends a response containing the requested resource or an error message.

Step 5: Browser Displays Result

The browser interprets the response and displays the web page to the user.

Simple Flow

User → Web Browser → Web Server → Response → Web Browser → User

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1.3 CLIENT-SIDE PROGRAMMING

Definition

Client-side programming refers to programming in which the code is executed on the user's device, usually inside a web browser.

It is mainly used to create:

- Interactive web pages
- Responsive interfaces
- Dynamic content
- Form validation
- Animations
- User interactions

Client-side programming allows many operations to happen directly in the browser without sending every action to the server.

Main Client-Side Technologies

1. HTML

HTML stands for **HyperText Markup Language**.

It is used to define the structure and content of a web page.

HTML is used to create elements such as:

- Headings
- Paragraphs
- Images
- Links
- Tables
- Forms
- Lists

2. CSS

CSS stands for **Cascading Style Sheets**.

It is used to control the appearance and presentation of web pages.

CSS controls:

- Colors
- Fonts
- Layout
- Spacing
- Backgrounds
- Borders
- Responsive design

3. JavaScript

JavaScript is a programming language used to add logic and interaction to web pages.

It can be used for:

- Form validation
- Buttons and events
- Animations
- Dynamic content
- Calculations
- Interactive menus
- Updating page content

Advantages of Client-Side Programming

1. Faster Interaction

Many operations happen directly in the browser, which can provide faster responses.

2. Reduces Server Requests

Client-side processing can reduce unnecessary communication with the server.

3. Improves User Experience

Interactive interfaces make websites easier and more enjoyable to use.

4. Form Validation

Client-side programming can check form data before it is sent to the server.

5. Dynamic Web Pages

It allows web page content and appearance to change dynamically.

Limitations of Client-Side Programming

1. Depends on Browser

The execution of client-side code depends on the user's browser and device.

2. Code Can Be Inspected

Client-side code can generally be viewed or inspected by users. Therefore, sensitive information should not be stored in client-side code.

3. Security Restrictions

Browsers apply security restrictions that limit access to certain local resources.

4. Browser Compatibility

Different browsers or environments may sometimes behave differently, so developers need to test their websites properly.

Client-Side vs Server-Side Programming

Client-Side Programming	Server-Side Programming
Runs in the user's browser.	Runs on the server.
Common technologies include HTML, CSS, and JavaScript.	Technologies include PHP, Python, Java, Node.js, ASP.NET, etc.
Mainly used for interface and user interaction.	Used for business logic and server processing.
Can perform client-side validation.	Handles databases and authentication.
Results are generally displayed directly in the browser.	Generates or provides data/content to the client.

1.4 IDE APPLICATIONS OF HTML

What is an IDE?

IDE stands for **Integrated Development Environment**.

An IDE is a software application that provides a collection of tools in a single environment for:

- Writing code
- Editing code

- Testing code
- Debugging code
- Managing projects

For HTML development, IDEs and code editors help developers create, edit, organize, and maintain web pages efficiently.

Common IDEs and Editors Used for HTML

1. Visual Studio Code

Visual Studio Code is a popular source-code editor used for web development.

Features include:

- Syntax highlighting
- Code completion
- Extensions
- Debugging support
- Integrated terminal
- Git integration

2. Adobe Dreamweaver

Adobe Dreamweaver is a web development tool that provides both visual and code-based editing capabilities.

3. Sublime Text

Sublime Text is a lightweight and fast code editor commonly used for HTML, CSS, JavaScript, and other programming languages.

4. Notepad++

Notepad++ is a simple Windows-based source-code editor that can be used to create HTML documents.

5. WebStorm

WebStorm is a professional IDE mainly focused on JavaScript and modern web development.

6. Atom

Atom was a customizable source-code editor that was historically used for web development.

Applications/Uses of IDEs for HTML

1. Writing HTML Documents

IDEs allow developers to create and edit HTML files.

2. Syntax Highlighting

Different HTML tags, attributes, and values are displayed using different colors or formatting, making code easier to understand.

3. Auto-Completion

IDEs can automatically suggest HTML tags and code while typing.

4. Error Detection

Many IDEs help developers identify syntax errors and coding problems.

5. Project Management

Developers can manage HTML, CSS, JavaScript, images, and other files in a single project.

6. Browser Preview

HTML pages can be opened or previewed in a web browser to check the output.

7. Extensions and Plugins

Additional functionality can be added using extensions and plugins.

8. Version Control

Many modern IDEs support version control systems such as Git.

9. Increased Productivity

Features such as search, shortcuts, code completion, refactoring, and project management help developers work more efficiently.

Basic HTML Document Structure

```
<!DOCTYPE html>
<html>
  <head>
    <title>My Web Page</title>
  </head>

  <body>
    <h1>Hello World</h1>
  </body>
</html>
```

Explanation

Declares that the document is an HTML5 document.

The root element of the HTML document.

Contains information about the document, such as the title and metadata.

Defines the title displayed in the browser tab.

Contains the visible content of the web page.

Defines a main heading.

An HTML file can be created using an IDE or code editor and then opened in a web browser to view the rendered result.

1.5 WEB PROTOCOLS – HTTP AND FTP

What is a Web Protocol?

A **web protocol** is a set of rules that defines how computers, applications, clients, and servers communicate with each other over a network.

Two important protocols related to Web and Internet services are:

1. HTTP
2. FTP

HTTP – HyperText Transfer Protocol

Definition

HTTP stands for **HyperText Transfer Protocol**.

It is an application-layer protocol used for transferring web resources between clients, such as web browsers, and servers.

HTTP follows a **request-response communication model**.

Basic HTTP Communication

Step 1: Client Sends Request

The web browser sends an HTTP request to the server.

Step 2: Server Processes Request

The web server receives and processes the request.

Step 3: Server Sends Response

The server sends an HTTP response to the client.

Step 4: Browser Processes Response

The browser uses the response to display or process the requested resource.

Simple Flow

Client/Browser → HTTP Request → Web Server → HTTP Response → Client/Browser

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Common HTTP Methods

1. GET

The GET method is used to request data or a resource from a server.

Example: Requesting a web page.

2. POST

The POST method is used to send data to a server.

It is commonly used for:

- Submitting forms
- Sending login information
- Creating resources

3. PUT

The PUT method is generally used to create or replace a resource at a specified location.

4. PATCH

The PATCH method is used to partially modify an existing resource.

5. DELETE

The DELETE method is used to request the deletion of a resource.

HTTP Status Code Categories

HTTP responses contain status codes that indicate the result of a request.

Code Range	Meaning	Example
1xx	Informational responses	100 Continue
2xx	Successful requests	200 OK
3xx	Redirection	301 Moved Permanently

Code Range	Meaning	Example
4xx	Client errors	404 Not Found
5xx	Server errors	500 Internal Server Error

Common Status Codes

200 OK – Request was successful.

301 Moved Permanently – Resource has been permanently moved to another URL.

400 Bad Request – The server could not understand the request.

401 Unauthorized – Authentication is required.

403 Forbidden – The server refuses to allow access.

404 Not Found – The requested resource could not be found.

500 Internal Server Error – A general error occurred on the server.

HTTPS – HTTP Secure

HTTPS stands for **HyperText Transfer Protocol Secure**.

HTTPS is HTTP combined with security provided by **TLS (Transport Layer Security)**.

HTTPS helps protect communication between the browser and server by providing:

- Encryption
- Authentication
- Data integrity

It helps prevent unauthorized parties from easily reading or modifying information while it is being transmitted.

Common Ports

- HTTP – Port **80**
- HTTPS – Port **443**

FTP – File Transfer Protocol

Definition

FTP stands for **File Transfer Protocol**.

It is a network protocol designed for transferring files between a client and a server.

FTP is commonly used for:

- Uploading files

- Downloading files
- Managing remote files
- Transferring multiple files

Uses of FTP

1. Uploading Website Files

FTP can be used to upload HTML, CSS, JavaScript, images, and other website files to a hosting server.

2. Downloading Files

Users can download files from a remote server.

3. Managing Remote Files

Depending on permissions, FTP can be used to manage files and directories on a remote server.

4. Transferring Multiple Files

FTP is useful for transferring large numbers of files between systems.

Features of FTP

- Supports file uploading and downloading.
- Supports file and directory management operations depending on permissions.
- Traditional FTP uses separate control and data connections.
- Authentication may be performed using a username and password.
- Traditional FTP does not encrypt credentials or data.
- Secure alternatives such as **FTPS** and **SFTP** are used when encrypted file transfer is required.

FTP Port

Traditional FTP commonly uses **Port 21** for control communication. Data transfer uses additional connections and ports.

Difference Between HTTP and FTP

HTTP	FTP
HTTP stands for HyperText Transfer Protocol.	FTP stands for File Transfer Protocol.
Mainly used for transferring web resources and communicating with web applications.	Mainly used for transferring files between systems.

HTTP	FTP
Commonly used by web browsers.	Commonly used by FTP clients and file-transfer tools.
Uses a request-response communication model.	Designed around file-transfer operations.
HTTP commonly uses port 80.	Traditional FTP commonly uses port 21 for control.
HTTPS is the secure version of HTTP.	FTPS and SFTP are commonly used for secure file transfer.

UNIT I – QUICK REVISION

- **WWW:** World Wide Web; a global system of interconnected web resources accessed through the Internet.
- **Tim Berners-Lee:** Inventor of the World Wide Web.
- **HTML:** Used to create the structure of web pages.
- **Web Browser:** A client application that requests and displays web resources.
- **Web Server:** A system/software that stores, processes, and delivers web resources.
- **Client-Side Programming:** Programming code executed mainly in the user's web browser.
- **HTML:** Defines web page structure.
- **CSS:** Controls web page presentation and design.
- **JavaScript:** Adds logic and interaction to web pages.
- **IDE:** Integrated Development Environment used to write, manage, test, and develop code.
- **HTTP:** Protocol used for communication and resource transfer between web clients and servers.
- **HTTPS:** Secure version of HTTP using TLS protection.
- **FTP:** Protocol mainly used for transferring files between a client and server.

IMPORTANT EXAM QUESTIONS

1. Explain the history and development of the World Wide Web.
2. What is WWW? Explain its basic components.
3. Who invented the World Wide Web? Explain the important milestones in its history.
4. Define a Web Browser and explain its functions.

5. Define a Web Server and explain its functions.
6. Differentiate between a Web Browser and a Web Server.
7. Explain how a Web Browser and Web Server communicate with each other.
8. What is Client-Side Programming? Explain its advantages and limitations.
9. Explain HTML, CSS, and JavaScript as client-side technologies.
10. Differentiate between Client-Side and Server-Side Programming.
11. What is an IDE? Explain its applications in HTML development.
12. Explain the basic structure of an HTML document.
13. What is a Web Protocol? Explain HTTP and FTP.
14. What is HTTP? Explain the HTTP request-response cycle.
15. Explain the common HTTP methods GET, POST, PUT, PATCH, and DELETE.
16. Explain the different categories of HTTP status codes.
17. What is HTTPS? Explain its importance.
18. What is FTP? Explain its uses and features.
19. Differentiate between HTTP and FTP.
20. Write short notes on:

- WWW
- Web Browser
- Web Server
- Client-Side Programming
- IDE
- HTTP
- HTTPS
- FTP